1. Window functions: OVER

- GROUP BY squishes rows; windows keep every row
- OVER ( ) opens the window
- Empty brackets = the whole table

```
SELECT title, pages,
AVG(pages) OVER () AS avg_pages
FROM books;
```

title	pages	avg_pages
Harry Potter	309	249.5
Matilda	240	249.5
Wonder	310	249.5
The Hobbit	310	249.5
Holes	233	249.5

2. ROW_NUMBER, RANK, DENSE_RANK

- ROW_NUMBER: 1, 2, 3, no ties
- RANK: ties share, then skip
- DENSE_RANK: ties share, no skip

```
SELECT title, pages,
RANK() OVER (ORDER BY pages DESC) AS rnk
FROM books;
```

title	pages	rnk
Wonder	310	1
The Hobbit	310	1
Harry Potter	309	3
Matilda	240	4
Dog Man	240	4

3. PARTITION BY

- A separate race for each group
- Numbering restarts at 1 in every group
- Top 1 per group: wrap it and keep rn = 1

```
SELECT genre, title, RANK() OVER
(PARTITION BY genre ORDER BY pages DESC) AS rnk
FROM books;
```

genre	title	rnk
Classic	Charlotte's Web	1
Comedy	Matilda	1
Comedy	Dog Man	1
Drama	Wonder	1
Fantasy	The Hobbit	1

4. Running totals, LAG & LEAD

- SUM / COUNT OVER (ORDER BY ...) = running total
- LAG = previous row, LEAD = next row
- First / last row gives NULL

```
SELECT loan_date,
COUNT(*) OVER (ORDER BY loan_date) AS so_far
FROM loans;
```

loan_date	so_far
2026-01-05	1
2026-01-20	2
2026-02-10	3
2026-03-01	4
2026-04-02	5

5. Practice free + what's next

- sqliteonline.com, db-fiddle.com, sqlime.org (choose SQLite)
- sqlbolt.com for free lessons
- Paste the setup from the practice guide, click Run
- Next, Part 7: CTEs (WITH), recursive queries, EXISTS